

7. Sensory and motor areas of the cerebral cortex (II)

Motor cortex

CORTICAL CONTROL OF MOTOR FUNCTIONS

The voluntary movements are controlled by the motor areas of the CC :

- The upper motor neuron (UMN): formed by the motor cells of the CC and their axons that form the descending motor tracts (pyramidal and extrapyramidal) that terminate on the AHCs of spinal cord and motor nuclei of the cranial nerves in the brain stem.
- The lower motor neuron (LMN): formed of the axons of the AHCs (motor spinal nerves) and motor nuclei of the cranial nerves (motor cranial nerves, III, IV, V, VI, VII, IX, X, XI, XII) (figure 30).

THE MOTOR AREAS OF THE CEREBRAL CORTEX

1. The primary motor area (Area 4, pyramidal area)

Site: on the lateral surface of the cerebral hemisphere, it is located in the anterior wall of the central sulcus and adjacent portions of the precentral gyrus of the frontal lobe. On the medial surface, it occupies the anterior portion of the paracentral lobule.

In Motor Area 4: there is no IGL. The EPL and IPL are very large containing the large pyramidal Betz cells and so, it is called the pyramidal area.

N.B. :

- Commissural fibers: → connect corresponding points in the two cerebral hemisphere.
- Association fibers: → connect adjacent gyri in the same hemisphere.
- Projection fibers: → output fibers from the CC.

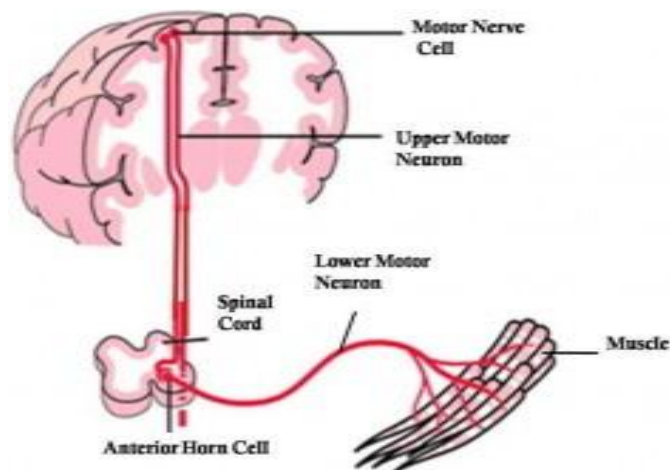


Figure 30: Upper and lower motor neurons

Connections Of Motor Area 4:

- Layers I and II: receive afferent fibers from the nonspecific thalamic nuclei and RAS → control the level of excitability of the CC (i.e. switch. it on and off).
- Layers III and IV: receive afferent fibers from:
 - The general somatic sensory areas, the auditory and visual cortical areas.
 - The PLVNT.
 - The VL and VA nuclei of thalamus, that carry information from the BG and cerebellum for coordination of motor functions.
 - The motor areas of the opposite side through the corpus callosum to coordinate motor functions of the corresponding points on either side of the body.
- Layers V and VI give the efferent projection motor fibers to:
 - AHCs and cranial nerve motor nuclei via the pyramidal tract (PT).
 - Subcortical motor centers via collaterals to: -Corpus striatum (CS).

-Red nucleus (RN). -Reticular formation (RF). -Inferior olivary and pontine nuclei.

- Inhibitory collaterals (lateral inhibition) to nearby cortical motor centers to localize the motor functions to the required group of muscles and prevent irradiation of impulses that produce undesired responses.

Representation of the body in area 4: (The motor homunculus) (figure 30)

- The representation is crossed.
- The representation is inverted.
- The greater and more complex the motor function of the part, the larger is the size of its area of representation.

The motor homunculus is a topographic representation of the body parts and its correspondents along the precentral gyrus of the frontal lobe. While the sensory homunculus is a topographic representation of the body parts along the postcentral gyrus of the parietal lobe (figure 31).

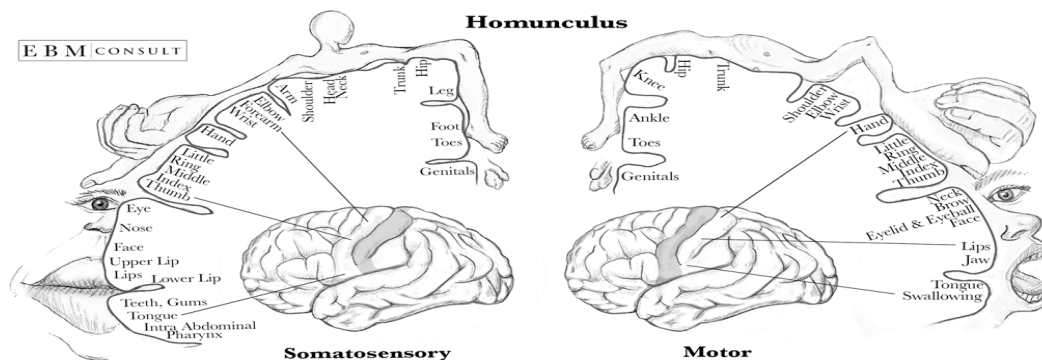


Figure 31 sensory and moto homunculus

Functions of area 4: It controls fine skilled movements performed by the distal joints.

- It was found that threshold stimulation of the finger or the thumb region of motor area 4 leads to contraction of small group of muscles of the finger,

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a single muscle or even a part of a muscle allowing for fine gradation of response as during playing the piano.

- It sends facilitatory impulses to the spinal centers of the stretch reflex.

2. The premotor area (Area 6):

Site: it lies in front of the primary motor area 4 in the frontal lobe. It lies both in the lateral and medial surface.

Histologically: it similar to area 4, but does not contain the large pyramidal Betz cells.

Functions:

- I. It is responsible for gross movements performed by proximal joints on the opposite side of the body.
- II. It contains specific functional areas areas:
 1. Broca's area for speech (Word formation area):
 - It lies anterior to area 4 and above Sylvian fissure.
 - Its stimulation produces movements of the lips, tongue, pharynx and larynx with spoken vocalization (i.e. the program of pronunciation).
 2. Hand skill area:
 - It lies anterior to motor area for the hands and fingers.
 - It is the memory center for voluntary skilled movements of the hands, e.g. lighting cigarette.
 3. Eye movement area (Frontal eye field, area 8):
 - It lies above Broca's area.

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- Its stimulation produces conjugate deviation of both eyes to opposite side, as well as, autonomic effects related to vision as lachrymation, lid closure and opening, pupillodilatation or constriction.

4. Head rotation area:

- It lies above the eye movement area.
- Its stimulation produces turning of the head to the opposite side.

III. Autonomic functions: area 6 controls selective autonomic functions related to somatic motor activity:

1. Stimulation of Broca's area → salivation not related to digestion but to facilitate articulate speech.
2. Stimulation of hand skill area → psychic sweating of the hand not related to temperature regulation, but to facilitate firm grip.
3. Stimulation of the eye movement area → autonomic effects related to vision as lachrymation to lubricate the eye ball, lid closure and opening to spread tears, pupillodilatation or constriction.
4. Cardiovascular responses on opposite side irrespective of somatic motor response:
 - I. Stimulation of certain sites of area 6 → VC ,
↑ABP, tachycardiaetc.
 - II. Stimulation of other sites in area 6 produces the opposite.

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IV. Inhibition of: the stretch reflex- the Grasp reflex.

Connections of area 6:

- Subcortical connection with area 4.
- Subcortical connection with sensory areas.
- With thalamic motor nuclei for coordination of movements.
- With BG and extrapyramidal motor centers of the brain stem.

The Pyramidal Tracts

- They are pathways for voluntary movements.
- They include:

1. Corticospinal Tract (CST) (figure 32):

Origin:

- It arises mainly from area 4 except the face region, from area 6 and from the sensory area as follows:
- Each tract contains about 1 million fibers:
- 3% of the fibers are axons of the large pyramidal Betz cells of layer V of area 4
- 60% of the fibers are small myelinated fibers that arise from other layers of area 4, premotor area 6 and primary and somatic association areas.
- The remaining fibers are thin non myelinated fibers.

Pathway:

- The fibers descend in the corona radiata (CR) → to:
- The internal capsule (IC); where the fibers occupy the genu and anterior

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2/3 of the posterior limb intermingled with extrapyramidal tract → to

- The MB, where the fibers occupy the intermediate 3/5 of the cerebral peduncle → to
- The pons, where the fibers descend in the bundles separated by the transverse pontine fibers that form the middle cerebellar peduncle → to
- The MO, where the fibers collect to form the Pyramid in the upper medulla → to lower medulla where:
- 80% of the fibers cross to the opposite side to form the Pyramidal Decussation then descend in the lateral white matter of the spinal cord forming the LCST (crossed) to end on the IN in the dorsal horn gray matter before relaying on the AHCs.
- 20% of the fibers descend without crossing in the anterior white matter of the spinal cord forming the direct ACST that gradually and partially crosses in the spinal cord to end on the AHCs of the same and opposite sides.

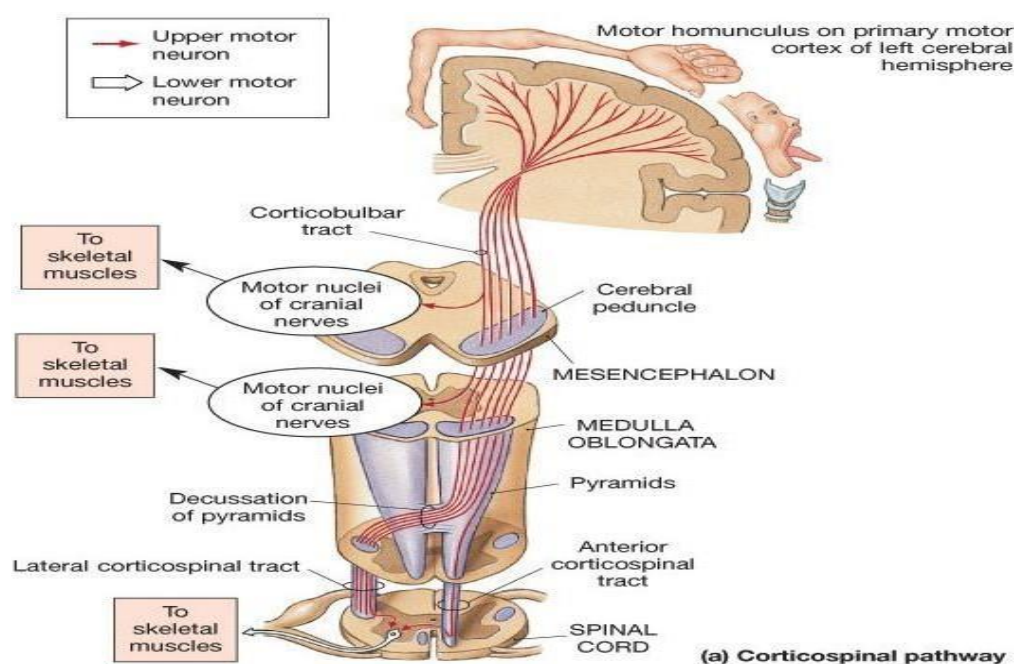


Figure 32: corticospinal and corticobulbar tracts

Functions of CST : nerves (3,4,6) of both sides, which control movements of the extraocular muscles.

Functions :

- It initiates and controls conjugate movements of both eyes to the opposite side.
- It mediates accommodation for near vision.

The Extrapyramidal Tracts:

Definition:

- These are the descending tracts other than the pyramidal that do not occupy the pyramid in the medulla.

Functions:

- It controls gross postural movements performed by proximal joints that provides postural background for performing fine skilled movements e.g. putting the hand in the writing position.
- It controls gross postural movements e.g. swinging the arms during waking.
- Some tracts are facilitatory and others are inhibitory and the resultant is predominant inhibition on the lower motor neuron centers.
- Responsible for appropriate distribution of muscle tone and postural reflexes.
- Mediate impulses from vital centers of the BS to spinal cord centers e.g. control of respiratory and VM reflexes.
- Origin: From area 6 -From area 4s. -From sensory areas.
- Pathway:

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The fibers descend in the CR and IC intermingled with the PT.

The fibers reach the BG : Caudate → to Putamen → to Globus Pallidus → general discharging center to :

Thalamus, sub thalamus, and SN → feedback control control to CC.

RN → spinal cord via rubrospinal tract.

Superior and inferior colliculi of MB → to spinal cord via Tectospinal Tracts:

The superior colliculus relays visual impulses via the lateral Tectospinal tract → to AHCs and LHCs of T1 and T2 to mediate visuo-spinal reflexes:

Reflex raising the hands to protect the eye.

Reflex movement in response to visual stimuli.

Reflex dilatation of the pupil in response to painful stimuli.

The inferior colliculus relays auditory impulses via the ventral Tectospinal tract → to spinal cord motor centers to mediate the following reflexes:

Reflex turning of the head in response to sound.

Reflex changing in position e.g. running on hearing a sound.

RF → to spinal cord via reticulospinal tracts.

Inferior olivary nucleus → to spinal cord via the Olivospinal tract.

The vestibular nuclei → to spinal cord via Vestibulospinal tract.

The descending autonomic tracts associated with the extrapyramidal tracts:

The respiratory tracts: from the RC in medulla → through the anterior and outer part of the lateral column of spinal cord → to AHCs supplying the diaphragm and intercostal muscles.

The vasomotor tracts: from the VMC in the MO → descend anterior to LCST → to LHCs of the thoracolumbar region (sympathetic spinal centers).

The sweat secretion tracts: from the heat regulating center in the hypothalamus → descends anterior to LCST → to LHCs of the thoracolumbar region (sympathetic spinal centers).

The vesicomotor tracts: both vesico motor and sensory tracts for voluntary control of micturition lie lateral and posterior to LCST. They partially decussate in the sacral region so that the sacral centers for micturition are bilaterally controlled. Micturition is not affected in hemisection of the cord.

CLINICAL APPLICATIONS (NERVOUS LESIONS)

(1) Removal of area 4:

- The manifestations occur in the opposite side.
- After a stage of spinal shock, the reflexes recover gradually to show the following responses on the opposite side of the body:
- Paralysis of the muscles characterized by:
 1. Being incomplete, due to the ipsilateral corticospinal and extrapyramidal tracts.
 2. Permanent loss of fine delicate movements performed by distal joints of fingers.
- The superficial reflexes are lost. The planter reflex shows +ve Babiniski sign in the form of dorsiflexion of the big toe without fanning of the lateral 4 toes which show normal planter flexion.
- Flaccidity: due to interruption of the supraspinal facilitatory pyramidal tract on the affected side.

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- The deep reflexes are depressed i.e. :
 1. Their threshold is high.
 2. Become slow or difficult to obtain.

(2) Removal of area 6:

The manifestations occur in the opposite side of the body:

- There is paresis, but not paralysis.
- Apraxia: inability to do purposeful movements by the affected muscles although the muscles are not paralyzed:
 1. Inability to perform fine skilled movements.
 2. Motor aphasia.
 3. Loss of voluntary turning of the head or conjugate deviation of the eyes to the opposite side.
- The planter response shows +ve Babiniski sign in the form of dorsiflexion and fanning of the lateral 4 toes only.
- Due to interruption of the supraspinal inhibitory impulses on the stretch reflex mediated by the extrapyramidal system we get: - Spasticity. - Hyperreflexia. - Clonus.
- The grasp reflex reappears:
- Autonomic disturbances in the affected side as VD, increased temperature and blood flow.

(3) Lesions of IC (figure 33):

Anatomically:

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- It is a V-shaped mass of white nerve fibers formed of anterior limb, genu and posterior limb.
- Present between the caudate nucleus and thalamus medially and the putamen and globus Pallidus laterally.
- The anterior limb of internal capsule (or frontal part) contains:
 1. fibers running from the thalamus to the frontal lobe
 2. fibers connecting the lentiform and caudate nuclei
 3. fibers connecting the cortex with the corpus striatum
 4. fibers passing from the frontal lobe through the medial fifth of the base of the cerebral peduncle to the nuclei pontis
 5. thalami pontine fibers
- The arrangement of the fibers in the IC is as follows:
 1. The pyramidal with the intermingled extrapyramidal tracts occupy the genu and the anterior 2/3 of the posterior limb.
 2. The sensory radiation lies immediately behind the pyramidal in the posterior limb.
 3. The optic radiation lies next.
 4. The auditory radiation lies most posteriorly.

Manifestations of ICL:

- UMNL occur on the opposite side of the body characterized by:
 1. Hemiplegia 2. Hemianaesthesia 3. Homonymous hemianopia
 2. Hearing diminution

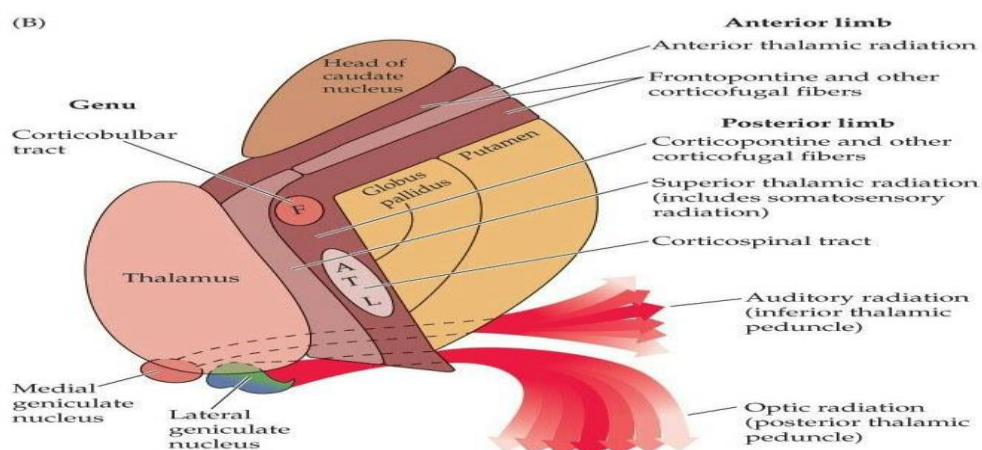


Figure 33 Internal capsule

UMNL and LMNL (table 13)

Table 13: comparison between UMNL and LMNL:

	UMNL	LMNL
Cause	Cerebrovascular strokes due to hemorrhage or thrombosis in the post limb of internal capsule ⇒ damage of both pyramidal and extrapyramidal fibers	1- Lesion of the lower motor neurons as in poliomyelitis 2. Damage of motor nerves e.g. DM or alcoholism
Characters		
1- Paralysis	* On the opposite side of the body (contralateral) * Widespread affecting half of the face, upper & lower limbs * Poor recovery	* On the same side of the lesion * Localized to muscles supplied by the affected segment only * Recovery may occur.

	UMNL	LMNL
2- Muscle tone	* Hypertonia of the spastic type in the paralyzed muscle * <u>Klasp</u> knife type: resistance to passive movement then sudden release * Cause: loss of inhibitory effect of the cortical extrapyramidal area & ↑↑ facilitatory impulses on the γ motor neurons $\Rightarrow$ facilitation of stretch reflex	* Hypotonia or <u>Atonia</u>: Flaccid paralysis (loss of tone in paralyzed muscles) * Cause: interruption of stretch reflex
3- Deep reflexes	* Exaggerated deep reflexes on the affected side: (e.g. knee & ankle jerks). * Clonus is present. * Cause: release of stretch reflex from cerebral inhibition	* Absent deep reflexes in muscles supplied by the affected segments or motor nerves

	UMNL	LMNL
4- Superficial reflex	* Lost on the affected side. * Cause: loss of supra-spinal facilitation * Abdominal & <u>ceremasteric</u> reflexes: absent * The planter reflex $\Rightarrow$ +ve <u>Babiniski's</u> sign.	* Lost on the affected segments only
5- Muscle wasting	* Not significant * Cause: paralyzed muscles are still innervated and can contract reflexly. * Spasticity saves muscle from wasting.	* Marked (disuse atrophy) * Cause: muscles cannot contract neither reflexly nor voluntary
6- Fasciculations	Absent	Present Visible spontaneous contractions of bundles of fibers in the affected ms

	UMNL	LMNL
7- Electric excitability	Normal a. Faradic current $\Rightarrow$ tetanic contractions b. Galvanic current $\Rightarrow$ normal response i) Chronaxie: normal ii) cathodal closing contraction is stronger than anodal closing contraction (CCC > ACC)	Shows changes a. Faradic current $\Rightarrow$ no response b. Galvanic current $\Rightarrow$ reactions of degeneration i) Chronaxie: prolonged ii) ACC > CCC